

NEW PRODUCTS

MAKE IN INDIA

COMPONENTS

LDO for automotive applications

AP7387Q from Diodes is a linear voltage regulator suitable for harsh automotive applications. It can work with a wide range of input voltage of 5V to



60V and typically 700mV at 100mA output current.

It features a

very fast line/load transient response against the rapid input voltage and load current changes. The IC consists of a voltage reference, an error amplifier, a resistor network for setting output voltage, a current limit circuit for current protection, short-circuit and thermal shutdown protection. The AP7387Q is a low-quiescent-current, high-PSRR linear regulator able to drive maximum of 150mA output current. The LDO is AEC-Q100 Grade 1 qualified, manufactured in IATF 16949 certified facilities, and supports PPAP documentation. It is suitable for use in various automotive applications, including automotive audio, navigation, tail lights, transmission control units, and battery management systems.

Diodes Incorporated

<https://www.diodes.com>

Application-specific MOSFETs

PSMN1R9-100SSE and PSMN2R3-100SSE from Nexperia are two new



application-specific MOSFETs (ASFETs) rated for 80V and 100V operation, respectively. The

ASFETs offer 40% lower RDS(on) and an increased power density of 58x for telecom and hot-swap computing ap-

plications. These MOSFETs come in a smaller size compared to any existing product with similar rating, making it possible for designers to create smaller modules for hot-swap applications.

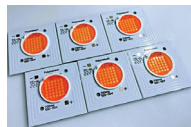
They come in 8mm × 8mm LPAK88 package that is suitable for use in high temperatures applications. The company has simplified the job of design engineers by providing hot SOA curves in the datasheet. The low RDS(on) offered by the MOSFETs helps in minimising the I²R losses when the ASFET is fully turned on. The MOSFETs achieve 10% SOA improvement compared to previous generations in D2PAK packages.

Nexperia

<https://www.nexperia.com>

LEDs for smart horticulture

Polymatech Electronics has released Ravaye, a series of LEDs for smart horticulture. The monochromatic



LEDs can enhance greenhouse and vertical farm lighting for professional horticulture

applications. They produce a range of wavelength that can promote healthier plant growth by accelerating photosynthesis, strengthening plant immunity, and increasing nutritional value. The Ravaye LEDs offer a wide variety of wavelength combinations to suit different plant needs. Their light can enhance plant growth, improve farming environments, and help lower a farmer's lighting system cost, thus increasing profitability for the farmers.

Polymatech Electronics

www.polymatech.in

SoC featuring ADAS and DMS

This system on chip (SoC) combines advanced driver assistance system (ADAS) with driver and occupant monitoring. The SoC won't only reduce the size of the component but will also lower the cost, complexity, and speed of adding advanced safety



systems to a broad range of vehicles.

This integrated solution will require

low power consumption, reducing thermal-management needs, which is critical for electric vehicles. The low power and high performance capability enable cameras to be packaged in a single box that can be placed in high-temperature areas, thus giving freedom to designers to place the system anywhere in the vehicle. This allows designers to preserve their development investments while tailoring the selected SoC's processing power to the needs of each car model's features, from basic ADAS and driver monitoring system (DMS) to full domain control using the 5nm CV3 family for level 2+ to L4 autonomy and beyond.

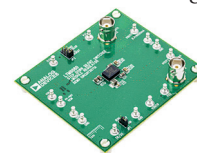
Seeing Machine, <https://seeingmachines.com>

Autobrain, <https://www.autobrain.ai>

Ambarella, <https://www.ambarella.com>

Low-noise dual-output regulator

LTM8080 is an ultra-low-noise dual-output µModule (micro module) regulator from Analog Devices. To reduce



the electromagnetic interference (EMI) emissions, the module comes enclosed in a metal-

lic EMI shield and employs 'Silent Switcher' architecture. The regulator can have an input voltage of up to 40V